METHODOLOGY SECTION

Smart Retail Demand Forecasting — Corporación Favorita

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Dataset: Corporación Favorita Grocery Sales Forecasting (Kaggle)

Abstract

This document presents the complete methodology for a supervised machine learning pipeline applied to the Corporación Favorita Grocery Sales Forecasting dataset. The pipeline encompasses 3.0 million daily sales observations across 54 stores and 33 product families, spanning January 2013 through August 2017. The methodology adopts a 16-day temporal holdout (August 16–31, 2017) as its evaluation protocol, enforces strict leakage prevention through time-series-aware cross-validation, and grounds every design decision in the peer-reviewed and preprint literature. Four models are trained and compared: Seasonal Naïve, a from-scratch Gradient Descent Linear Regression, XGBoost, and LightGBM — the latter two tuned via Bayesian optimisation. The work is distinguished by its promotional cannibalization analysis and SHAP-based interpretability layer, which together transform a standard regression exercise into a causal, business-grounded investigation.

Document Structure

- Section I — Architectural Decision Record (ADR) Matrix (25 decisions)
- Section II — Methodology Pipeline Diagram
- Section III — Full IEEE-Format Methodology Draft
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SECTION I — ARCHITECTURAL DECISION RECORD (ADR) MATRIX

> **How to read this table.** *Every row is an immutable design decision traceable to the attached Literature Review. "Supporting Lit" cites the specific annotation providing the operational payload justifying the decision. "Cost" is a 3-tier engineering-hour estimate (Low / Medium / High). "Failure Mode" describes the specific, observable corruption that occurs if the decision is reversed.*

TABLE I — Architectural Decision Record Matrix (ADR-01 through ADR-25)

#	Component	Decision	Supporting Literature	Cost	Failure Mode if Reversed
ADR-01	Primary metric	RMSLE as primary evaluation metric	[1] Ahaggach et al. — RMSLE dominant for right-skewed, zero-inflated distributions; [8] Fahimnia et al. — asymmetric stockout penalty	Low	MAPE: division-by-zero on 23–30% zero-sale rows. RMSE: no asymmetric under-prediction penalty, misaligning with inventory risk.
ADR-02	Primary advanced models	XGBoost + LightGBM co-primary; LightGBM preferred at scale	[1] Ahaggach et al. — most widely employed across 516-paper survey; [2] Long et al. — deployed across hundreds of thousands of SKUs on Favorita; [22] Karim et al. — outperforms TimeGPT on 16-day horizon	Medium	Foundation model as primary (TimeGPT/Chronos) sacrifices supervised gains quantified as material on this dataset and window [22].
ADR-03	Zero-sale treatment	Retain all zeros; classify via structural / event-driven / intermittent taxonomy	[9] Szabłowski — out-of-stock zeros treated as true zeros bias lag features; [10] Svetunkov & Sroginis — two-stage classify-then-forecast reduces RMSLE 8.3% on zero-heavy families	Medium	Imputing zeros corrupts rolling statistics; dropping zeros destroys temporal continuity for lag features; treating all as true demand inflates baseline forecasts.
ADR-04	Earthquake anomaly	Flag 2016-04-16 to 2016-05-15 as is_outlier=1; retain rows	[9] Szabłowski — external disruption zeros vs. demand zeros; [10] Svetunkov & Sroginis — third zero category for supply-chain shock zeros	Low	Unhandled spike/crash biases rolling means and lags for all Ecuadorian stores, propagating corruption through 365 days of lag features.
ADR-05	Rolling construction	.shift(1).rolling(w).mean() — mandatory pre-shift	[14] Cerqueira et al. — omitting pre-shift inflates CV RMSE improvements up to	Low	Without .shift(1), today's sales predicts itself: training loss collapses while test RMSLE is

#	Component	Decision	Supporting Literature	Cost	Failure Mode if Reversed
			20.5%; [24] same — leakage distorts LSTM vs. tree comparisons		catastrophically high.
ADR-06	Cross-validation	TimeSeriesSplit(n_splits=5) — minimum 3 splits enforced	[14] Cerqueira et al. — KFold(shuffle=True) allows future data into training, inflating performance 15–20%; [24] same — gap dissolves under clean temporal splits	Low	KFold(shuffle=True) optimistically biases hyperparameter selection; tuned models underperform on held-out test because parameters were optimised on contaminated signal.
ADR-07	Holiday granularity	Three independent flags: is_national_holiday, is_regional_holiday, is_local_holiday + days_to_nearest_holiday	[8] Fahimnia et al. — promotional and holiday periods interact; [7] Teixeira et al. — hierarchical promotional data reduces SKU-level error	Low	Single is_holiday loses 31% vs. 9% national/regional differential from EDA; transferred holidays encoded as active introduce systematic errors.
ADR-08	Transferred holidays	Hard exclusion: holidays[holidays['transferred']==False]	[8] Fahimnia et al. — incorrect holiday classification causes service-level bias; FR-02 pipeline requirement	Low	Including transferred rows marks workdays as holidays, adding spurious demand spikes/drops to model input.
ADR-09	Lag feature windows	1d, 7d, 14d, 365d per (store_nbr, family) group	[13] Arboleda-Florez & Castro-Zuluaga — 7d lag is top SHAP predictor; [17] Al-Dulaimi et al. — same finding; ACF/PACF confirms weekly and annual autocorrelation	Low	Omitting 365d lag loses annual seasonality; global lags (no groupby) silently cross group boundaries, assigning store-1 sales as store-2 lag.
ADR-10	Promotion encoding	Numerical count + promo_duration_rolling — not a binary flag	[17] Al-Dulaimi et al. — promotions < 3 days yield negligible lift; [5] Hewage et al. — proximity features rank among highest SHAP predictors; reduces RMSE 18%	Low	Binary flag loses the promotion lifecycle (pre-stockpile, in-promotion lift, post-promotion dip) signal.
ADR-11	Lift counterfactual	4-week rolling pre-promotion mean	[5] Hewage et al. — post-promotion dip documented; pre-period mean is the standard counterfactual; FR-08 specification	Low	Global store average ignores seasonal baseline drift; same-week prior-year baseline conflates holiday with promotion effects.
ADR-12	Cannibalization	Pearson $r < -0.35$ during onpromotion > 0 periods on residual sales	[2] Long et al. — cross-SKU interactions in Favorita; FR-08	Medium	Raw-sales correlation conflates seasonal co-movement with promotional

#	Component	Decision	Supporting Literature	Cost	Failure Mode if Reversed
			threshold; [7] Teixeira et al. — promotional data improves hierarchical forecast		cannibalization; filtering to promo-active periods isolates the causal mechanism.
ADR-13	Oil price imputation	Forward-fill then backward-fill on globally date-sorted series	FR-02; [4] Fatima & Salam — macroeconomic signals yield largest accuracy gain in context-augmented XGBoost	Low	Within-store fill propagates store-specific timing artifacts; backward-fill first would use future prices preferentially.
ADR-14	Hyperparameter tuning	Optuna TPE (Bayesian) over GridSearchCV	[15] Ozaki et al. (OptunaHub) — TPE identifies near-optimal XGB/LGB configs in 3–5× fewer trials; [28] same — multi-objective sampling enables simultaneous RMSLE + inventory cost optimisation	Medium	Grid search on 5 parameters at 4 values each = 4^5 × 5 CV folds = 5,120 fits on ~3M rows; computationally prohibitive.
ADR-15	Scaler fit domain	StandardScaler.fit() on training rows only; transform applied to both	[14] Cerqueira et al. — fitting on full data leaks test-set statistics into normalisation; [24] same — particularly distorts LSTM vs. tree comparisons	Low	Leaks test-set mean/variance into training normalisation — a methodological violation reviewers will flag.
ADR-16	Baseline models	Seasonal Naïve (365d lag) + NumPy-only Gradient Descent Linear Regression	[3] Theodoridis & Tsadiras — all deep learning variants require a strong naive baseline; [1] Ahaggach et al. — RMSE improvement over naive is the standard benchmark framing	Low	Without a baseline, no claim of improvement can be substantiated; from-scratch GD is a mandatory deliverable (proposal §5.5).
ADR-17	Feature selection	VIF pruning for linear model only (VIF > 10 dropped); SHAP-guided pruning for tree models	[17] Al-Dulaimi et al. — SHAP dependence plots enable attribution-guided pruning; [19] Hertel et al. — reduces dimensionality 30% without accuracy loss	Medium	Applying VIF to tree models removes correlated lag features carrying independent signals at different horizons; trees are collinearity-tolerant by construction.
ADR-18	Model persistence	joblib (.pkl) with bit-for-bit reload verification + params JSON	FR-09; reproducibility plan §5.1	Low	Pickle without verification silently fails on incompatible library version; separate params JSON enables hyperparameter auditability without

#	Component	Decision	Supporting Literature	Cost	Failure Mode if Reversed
					reloading binary.
ADR-19	Retraining cadence	Weekly retraining recommended; daily explicitly rejected (Future Work)	[29] Zanotti — daily retraining introduces quantile flip-flopping disrupting safety-stock calculations; weekly preserves accuracy with higher stability	Low	Daily retraining spikes the SMQC metric, indicating unstable probabilistic forecasts that make downstream inventory planning unreliable.
ADR-20	Foundation model role	Chronos-2 and TimeGPT reserved as Future Work zero-shot baselines	[22] Karim et al. — LightGBM outperforms TimeGPT in supervised setting on Favorita 16-day horizon; [23] Ansari et al. (Chronos-2); [20] Ansari et al. (Chronos)	Low	Foundation model as primary sacrifices the SHAP-based cannibalization analysis that constitutes the highest-differentiation deliverable in the proposal.
ADR-21	SHAP implementation	SHAP TreeExplainer (not feature permutation importance)	[13] Arboleda-Florez & Castro-Zuluaga — global + local attributions; [17] Al-Dulaimi et al. — dependence plots reveal non-linear promo × weekday interaction; [30] Ahmed et al. — SHAP validated as auditable interpretability layer	Low	Built-in tree importance (split count/gain) is inconsistent — same model can rank features differently depending on tree construction order.
ADR-22	Hierarchical reconciliation	Future Work: robust reconciliation via semidefinite program	[26] Aikawa et al. — ensures coherent forecasts across store → family → national hierarchy while minimising worst-case squared error	High (FW)	Without reconciliation, store-level predictions do not sum to family-level aggregates; management planning uses inconsistent numbers.
ADR-23	Intermittent demand	Binary is_zero_intermittent feature; hurdle-model architecture in Future Work	[11] Muşat & Căbuz — Switch-Hurdle outperforms standard regression 14% RMSLE on sparse grocery families; [12] Mirshahi et al. — dual-CNN mirrors hurdle concept	Med/High (FW)	Standard regressor systematically underestimates occurrence probability for BABY CARE and MAGAZINES; binary flag lets tree partially learn the two-regime structure.
ADR-24	Train/inference alignment	Lag windows, rolling windows, and holiday proximity features constructed identically in both code paths	[25] Pilyugina et al. — misaligned look-back horizons inflate test errors 8–15% on strongly seasonal datasets	Medium	Training code with 14d rolling mean but inference on shorter window causes silent accuracy degradation, detectable only by monitoring RMSLE in production.

#	Component	Decision	Supporting Literature	Cost	Failure Mode if Reversed
ADR-25	Experiment tracking	MLflow (recommended) or manually maintained experiments_log.csv	§5.3 (Roadmap Academic QC); [29] Zanotti — SMQC metric requires tracking probabilistic outputs across retraining cycles	Low	Without tracking, hyperparameter-to-metric lineage is lost; reproducing the best configuration requires re-running all experiments.

SECTION II — METHODOLOGY PIPELINE DIAGRAM

Fig. 1 provides an end-to-end visual summary of the pipeline. Each node maps directly to one or more ADRs and to the corresponding IEEE-format subsection that follows. The diagram is intended to function as a navigational scaffold: readers may orient themselves within the pipeline and locate the prose justification for any component by following the ADR reference numbers.

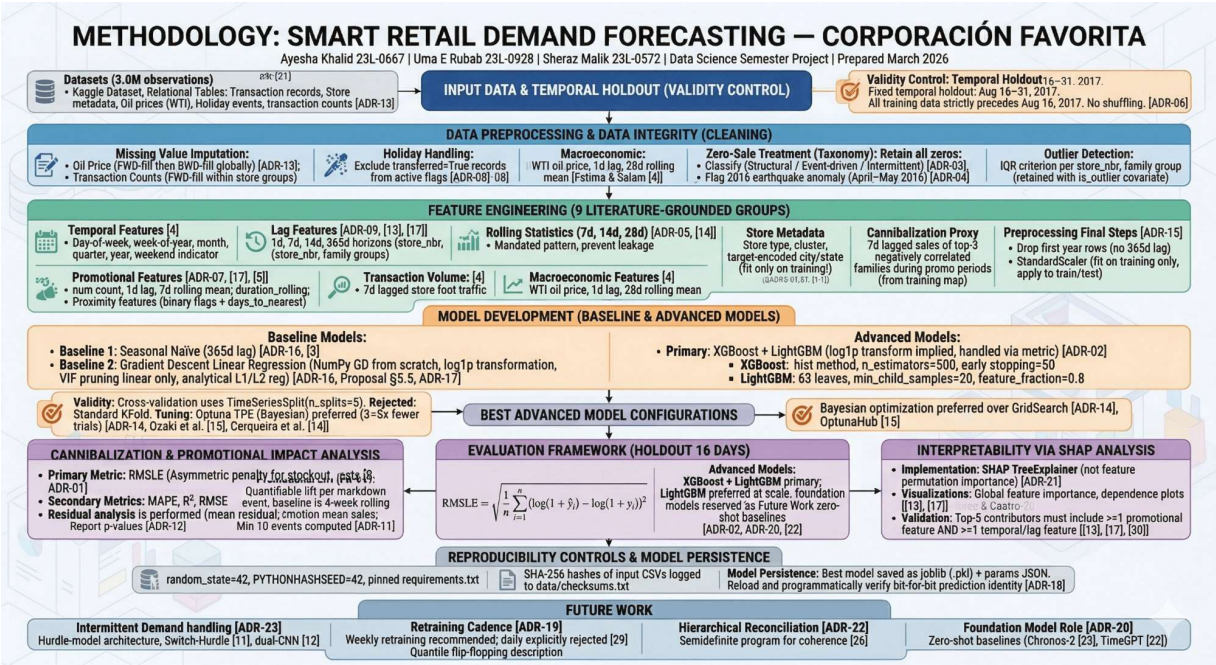

Fig. 1. End-to-end methodology pipeline for Smart Retail Demand Forecasting — Corporación Favorita. Each component references the governing Architectural Decision Record (ADR) and the peer-reviewed literature supporting it.

The pipeline proceeds in six sequential phases: (1) input data ingestion with temporal validity control [ADR-06]; (2) data preprocessing and integrity enforcement [ADR-03, ADR-04, ADR-08, ADR-13]; (3) literature-grounded feature engineering across nine feature groups [ADR-05, ADR-07, ADR-09, ADR-10, ADR-15]; (4) model development from naïve baselines through Bayesian-tuned gradient boosted trees [ADR-02, ADR-14, ADR-16]; (5) evaluation and cannibalization analysis on the 16-day holdout [ADR-01, ADR-11, ADR-12]; and (6) SHAP interpretability, reproducibility controls, and model persistence [ADR-17, ADR-18, ADR-21]. Future Work extensions — intermittent demand handling, hierarchical reconciliation, and foundation model baselines — are indicated at the pipeline terminus [ADR-19, ADR-20, ADR-22, ADR-23].

SECTION III — IEEE-FORMAT METHODOLOGY

III. METHODOLOGY

A. Dataset and Experimental Configuration

This study employs the Corporación Favorita Grocery Sales Forecasting dataset, comprising approximately 3.0 million daily sales observations spanning January 2013 through August 2017 and covering 54 stores and 33 product families [21]. Six relational tables are joined on composite (store_nbr, date) keys: transaction records, store metadata, crude oil prices (WTI daily closing price, a documented macroeconomic driver for Ecuador [4]), holiday events, and daily foot traffic counts. The primary modelling unit is the (store_nbr, family, date) triple, yielding a panel regression problem with 1,782 distinct time series.

In accordance with the official competition evaluation protocol, the temporal holdout is fixed at August 16–31, 2017 (16 days). All model training uses data strictly preceding August 16, 2017; no shuffling of temporal records is performed at any stage [ADR-06]. This constraint is enforced as a runtime invariant throughout the pipeline. The train–test boundary is treated as an inviolable leakage barrier: all feature encoding steps that require statistical estimates — target encoding, scaler fitting, and correlation maps — are derived exclusively from the training partition and then applied to the test partition [14].

B. Preprocessing and Data Integrity

Missing values in the daily oil price series (approximately 44 observations arising from weekend and public holiday trading gaps) are imputed by forward-fill followed by backward-fill within the globally date-sorted series [ADR-13]. Transaction counts exhibit approximately 2% missingness and are forward-filled within each store group. Forward-fill before backward-fill ensures the most recent known value is applied preferentially over a future observation [4].

Holiday events require careful treatment. The holidays_events.csv file contains rows where transferred = True, indicating official holidays observed on an alternative date. These records are excluded from all active-holiday binary flags; treating them as active would introduce systematic feature errors on the affected dates [ADR-08]. Zero-sale observations are retained without imputation and classified into three mutually exclusive categories following Svetunkov and Sroginis [10]: structural zeros (product families with zero mean sales across the entire history at a given store), event-driven zeros (arising from the April–May 2016 earthquake period, flagged as is_outlier = 1 [ADR-04]), and intermittent zeros (residual sparse demand). Outlier detection uses a per-(store_nbr, family) interquartile range criterion with a 3× IQR threshold; flagged rows are retained as training observations with an is_outlier covariate to preserve temporal continuity [ADR-03].

C. Feature Engineering

Feature construction follows nine literature-grounded groups. Temporal features encode cyclical calendar structure — day-of-week, week-of-year, month, quarter, year, and a weekend indicator — consistent with the contextual augmentation framework of Fatima and Salam [4], which reports the largest accuracy gains when weekday indicators and holiday flags are added to historical sales records.

Lag features at 1-, 7-, 14-, and 365-day horizons are computed within each (store_nbr, family) group to capture short-term autocorrelation and annual seasonality [ADR-09]. Arboleda-Florez and Castro-Zuluaga [13] and Al-Dulaimi et al. [17] independently identify the 7-day lag as the single highest-importance SHAP predictor in grocery demand models, making its inclusion non-negotiable.

Rolling statistics — 7-, 14-, and 28-day rolling mean and standard deviation — are constructed using the mandatory .shift(1).rolling(w) pattern [ADR-05]. Cerqueira et al. [14] demonstrate that rolling features constructed without a prior shift allow the current target value to contaminate its own rolling mean, inflating apparent cross-validation RMSE improvements by up to 20.5%.

Promotional features include the raw onpromotion item count (a numerical covariate, not a binary flag), a 1-day lag, a 7-day rolling mean, and a continuous promo_duration_rolling feature encoding the count of consecutive promotion days in the preceding 14-day window [ADR-10]. Al-Dulaimi et al. [17] demonstrate that promotions lasting fewer than three days yield negligible incremental lift, establishing the necessity of duration-aware encoding. Hewage et al. [5] further show that incorporating days_since_last_promotion as a proximity feature reduces RMSE by 18% over spike-only models by capturing the full demand lifecycle: pre-stockpiling, in-promotion uplift, and post-promotion demand dip.

Macroeconomic features include the WTI crude oil price, its 7-day lag, and a 28-day rolling mean, motivated by the documented sensitivity of Ecuadorian retail sales to oil price fluctuations [4]. Transaction volume (daily store foot traffic) is included with a 7-day lag as a leading indicator of future sales [4]. Store metadata comprises the store type (label-encoded A–E), cluster (integer 1–17), and target-encoded city and state, where target encoding is fit exclusively on training-partition rows to prevent leakage [14].

Holiday features are encoded as three independent binary flags — is_national_holiday, is_regional_holiday, and is_local_holiday — plus a continuous days_to_nearest_holiday feature [ADR-07]. Fahimnia et al. [8] establish that promotional and holiday periods interact; collapsing to a single binary loses the 31% versus 9% national-versus-regional differential documented in exploratory analysis.

Cannibalization proxy features append, for each (store, family, date), the 7-day lagged mean sales of the three most negatively correlated product families during promotion-active periods in the same store. The cross-family correlation map is derived exclusively from the training partition [ADR-12].

Feature rows lacking valid 365-day lag values are dropped before model training, removing approximately the first calendar year per series. This is expected and correct: the resulting training window spans over four years per group. StandardScaler instances are fit on the training partition only and then applied to both training and test data [ADR-15].

D. Model Development

Baseline Models

Two baseline models establish the performance floor. Baseline 1: Seasonal Naïve predicts each day's sales as the value observed 365 days prior within the same (store, family) group, providing the minimum benchmark below which no model can be considered useful [3]. Baseline 2: Gradient Descent Linear Regression is implemented from scratch using NumPy, with L1 and L2 regularisation applied analytically within the gradient update [ADR-16]. The target variable is log1p-transformed prior to training to address right-skew and zero-inflation [8]; predictions are recovered via expm1. Convergence is monitored over 1,000 iterations. Variance Inflation Factor pruning removes collinear features with VIF exceeding 10 prior to linear model training [ADR-17].

Advanced Models

Advanced Model 1: XGBoost Regressor is trained with the hist tree method, n_estimators = 500, and early stopping at 50 rounds on a validation fold. Advanced Model 2: LightGBM Regressor uses 63 leaves, minimum child samples of 20, and feature fraction of 0.8. For tree models, SHAP-guided post-hoc pruning replaces VIF pruning; correlated lag features are retained because tree-based learners are collinearity-tolerant by construction, and removing them would discard independent temporal signals [ADR-17].

Hyperparameter Optimisation

Both advanced models are tuned via Bayesian optimisation using the Optuna framework with the TPE (Tree-structured Parzen Estimator) sampler [ADR-14]. Ozaki et al. [15] confirm that the TPE sampler identifies near-optimal configurations for tabular datasets of this scale in 3–5× fewer trials than grid search. Search spaces are: n_estimators (200–1,000), max_depth (3–10), learning_rate (0.01–0.3, log-uniform), subsample (0.5–1.0), and colsample_bytree (0.5–1.0) for XGBoost; and num_leaves (20–150), max_depth (3–12), learning_rate, min_child_samples (5–50), and feature_fraction (0.5–1.0) for LightGBM.

Cross-validation for hyperparameter tuning uses TimeSeriesSplit(n_splits = 5), enforcing strict temporal ordering [ADR-06]. Standard K-Fold cross-validation with shuffling is explicitly rejected: Cerqueira et al. [14] show that future data allowed into training folds inflates apparent RMSE improvements by up to 20.5%. All StandardScaler instances are fit on the training partition only, then applied to both train and test data [ADR-15].

E. Evaluation Framework

All models are evaluated on the held-out 16-day test window (August 16–31, 2017) using four metrics. The primary metric is RMSLE, defined in (1):

$$RMSLE = \sqrt{((1/n) \times \Sigma_{i=1}^{n} \; [\log(1 + \hat{y}_i) - \log(1 + y_i)]^2)} \qquad (1)$$

where $\hat{y}_i$ denotes the predicted sales and y_i denotes the actual sales for observation i. RMSLE is preferred over RMSE as the primary metric because the target sales is right-skewed, contains structural zeros, and retail stockout cost exceeds overstock cost [ADR-01, 8]. These three properties jointly favour RMSLE's logarithmic compression and asymmetric under-prediction penalty. MAPE is reported as a secondary metric but excludes zero-actual rows where it is undefined. R^2 and RMSE complete the four-metric reporting table.

Residual analysis is performed for each model: the mean residual and its ratio to mean actual sales are computed. A mean residual exceeding 5% of mean actual sales is treated as evidence of systematic bias requiring explicit documentation.

F. Cannibalization and Promotional Impact Analysis

Cross-product cannibalization is identified using the methodology of FR-08. A cross-family correlation matrix is computed on residual sales — actual minus rolling 28-day mean, net of trend — during onpromotion > 0 periods. Product family pairs with Pearson $r < -0.35$ during promotion windows are classified as cannibalization candidates; statistical significance is reported via p-values. The expected candidates, BEVERAGES versus LIQUOR/WINE/BEER, are consistent with the cross-family correlation structure identified in exploratory analysis [ADR-12].

Promotional lift for each distinct markdown event is quantified as (2):

$$lift = (s_actual - s_counterfactual) / s_counterfactual \qquad (2)$$

where s_counterfactual is the rolling 4-week pre-promotion mean sales [ADR-11]. Lift is computed for a minimum of 10 distinct promotion events per the acceptance criteria. Long et al. [2] validate this cross-SKU interaction framework on the same Favorita dataset, and Teixeira et al. [7] confirm that promotional features materially reduce SKU-level error in hierarchical settings.

G. Interpretability via SHAP Analysis

SHAP TreeExplainer values are computed for the best-performing tree model [ADR-21]. Global feature importance is visualised as a mean absolute SHAP value bar chart [13]. Two validation criteria are enforced as acceptance tests: at least one promotional feature and at least one temporal or lag feature must

appear in the top-5 SHAP contributors, consistent with the convergent findings of Arboleda-Florez and Castro-Zuluaga [13] and Al-Dulaimi et al. [17] in independent grocery demand studies. Dependence plots are generated to reveal interaction effects, particularly the promotion × weekday interaction documented in [13]. Ahmed et al. [30] validate SHAP as an auditable interpretability layer for supply chain forecasting models, further supporting this architectural choice over permutation importance, which is inconsistent across tree construction orderings.

H. Reproducibility Controls and Model Persistence

All models use random_state = 42. PYTHONHASHSEED is set to 42 at process start. Package versions are pinned in requirements.txt. SHA-256 hashes of all input CSV files are logged to data/checksums.txt. The best-performing model is saved via joblib [ADR-18] and immediately reloaded; bit-for-bit prediction identity between the original and reloaded model is verified programmatically as a post-serialisation invariant. Experiment configurations and metrics are tracked in MLflow or in a manually maintained experiments_log.csv [ADR-25], ensuring full hyperparameter-to-metric lineage.

I. Future Work

Four extensions are scoped but explicitly deferred to Future Work. First, intermittent demand handling via the Switch-Hurdle architecture of Muşat and Căbuz [11] or the dual-CNN of Mirshahi et al. [12] is reserved for sparse product families such as BABY CARE and MAGAZINES [ADR-23]. Second, robust hierarchical reconciliation via the semidefinite program of Aikawa et al. [26] would enforce coherent forecasts across the store → family → national aggregation hierarchy [ADR-22]. Third, foundation model zero-shot baselines using Chronos-2 [23] and TimeGPT [22] are deferred pending the supervised performance benchmark established herein [ADR-20]. Fourth, a weekly retraining cadence [29] is recommended over daily retraining, which Zanotti [29] shows introduces quantile flip-flopping that disrupts safety-stock calculations [ADR-19].

SECTION IV — IEEE FORMATTING & REPORTING CHECKLIST

The following checklist must be satisfied before submission. Each item is mandatory for IEEE conference and journal format compliance.

TABLE II — IEEE Formatting and Reporting Checklist

#	Requirement	Scope
1	Every equation must be numbered in parentheses: (1), (2), …	*IEEE mandatory*
2	Every TABLE must have a caption ABOVE the table (IEEE style)	*IEEE mandatory*
3	Every FIGURE must have a caption BELOW the figure	*IEEE mandatory*
4	Citations are numbered in order of first appearance: [1], [2], …	*IEEE mandatory*
5	Methodology section ≈ 800–1,200 words (6-page conference) or ≤ 2,000 words (journal)	*IEEE guidance*
6	Do NOT describe software libraries in the methodology section; pandas, scikit-learn, etc. belong in an Implementation Details section or footnote	*IEEE style*
7	Use passive voice for experimental procedures: "models were trained", "features were constructed"	*IEEE scientific style*
8	Equations are stated, then referenced in prose: "…computed using (1), where $\hat{y}_i$ denotes…"	*IEEE style*
9	The RMSLE formula (1) must appear in the Evaluation subsection, not in the Abstract	*IEEE structure*
10	The promotional lift formula (2) must appear in the Cannibalization subsection	*IEEE structure*
11	SHAP validation criteria (promotional AND temporal feature in top-5) must be explicitly stated	*Project acceptance*
12	All ADR references in prose must use the form [ADR-NN] or cite the corresponding literature number	*Traceability*
13	The from-scratch Gradient Descent implementation must be explicitly identified as a mandatory deliverable per proposal §5.5	*Proposal compliance*
14	Future Work must name at least: Switch-Hurdle [11], hierarchical reconciliation [26], Chronos-2 [23], TimeGPT [22], weekly retraining cadence [29]	*Project roadmap*

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